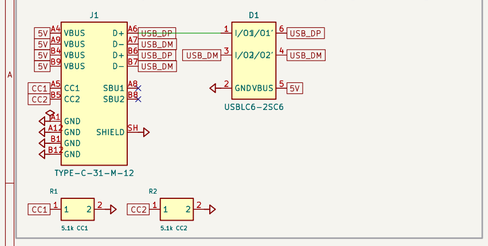
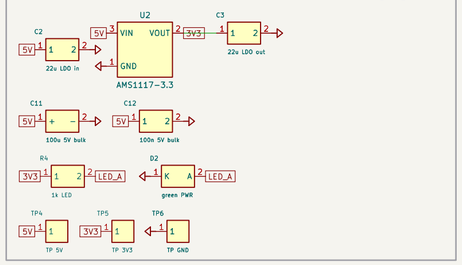
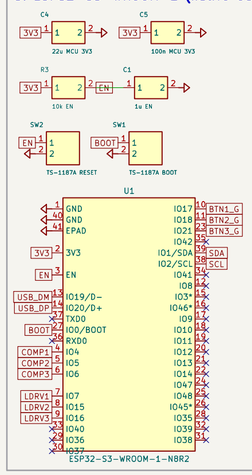
1. USB-C INPUT (UFP sink, 5V + native USB) [ADR-0001]



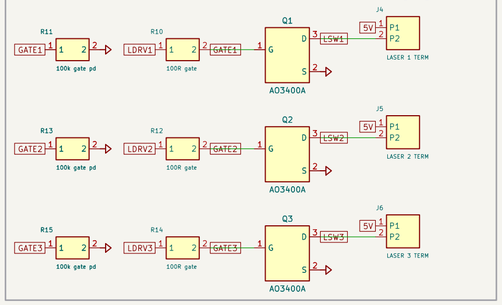
2. POWER: 5V -> AMS1117 -> 3V3; bulk at lasers [P4]



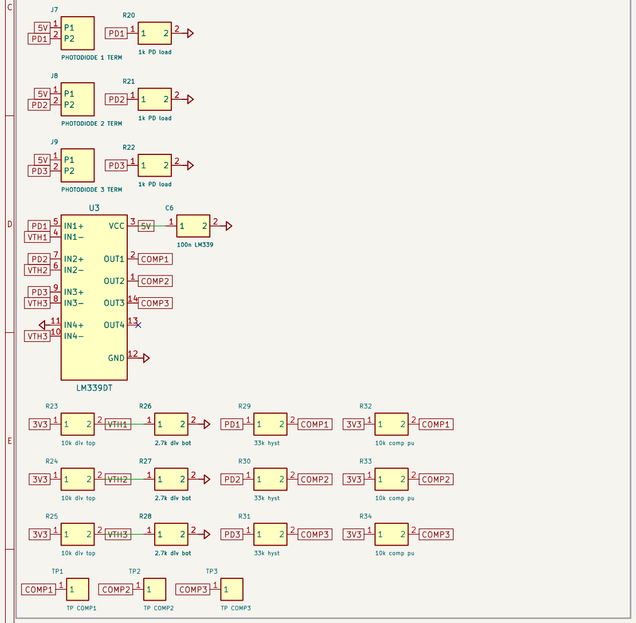
3. ESP32-S3-WROOM-1 (native USB; pin map ADR-0004)



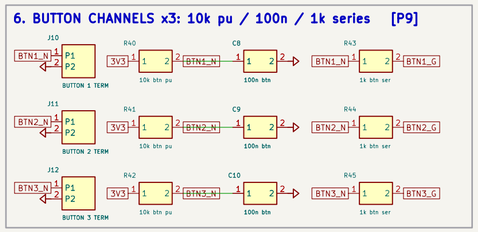
4. LASER CHANNELS x3: low-side AO3400A, off at boot [P5]



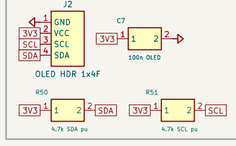
5. PHOTODIODE CHANNELS x3: BPW34 -> 1k load -> LM339 @5V [P6, ADR-0002]



6. BUTTON CHANNELS x3: 10k pu / 100n / 1k series [P9]



7. OLED HEADER (GND VCC SCL SDA — CHECK MODULE PINOUT) [P8]



Laser interruption timing bench controller: 01_docs/ + ADRs in repo

Sheet: /
File: esp32_laser_timing.kicad_sch

Title: esp32-laser-timing

Size: A2	Date: 2026-07-17	**Rev: v1.7**
KiCad E.D.A. 10.0.4		Id: 1/1